



# Fluent on Paper

Validating the mandatory AI-literacy self-assessment against measured task performance in a postgraduate cohort

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## Motivation

Every research student completes a mandatory AI-literacy module and self-certifies fluency across a fixed tool set. The sign-off feeds the induction dashboard as a capability measure. But does a confident self-assessment predict what a student can do at the keyboard?

## Research questions

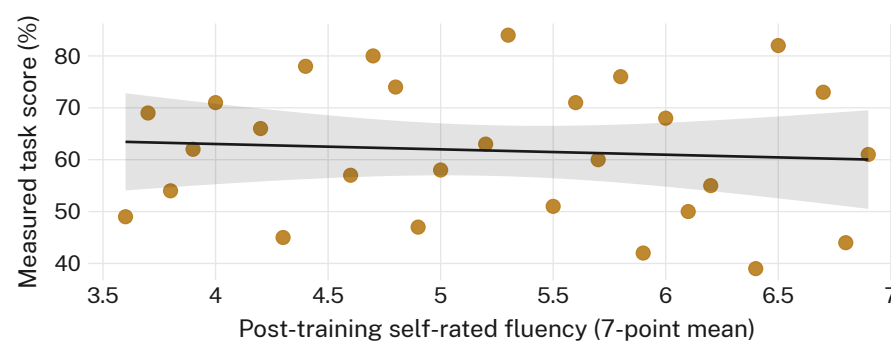
- Does post-module self-rated fluency predict measured task performance?
- Does the certification threshold separate competent from non-competent users?
- What does the sign-off track, if not skill?

## Study design

- $n = 118$  HDR students · 7-point self-rating + 14-task tool battery · single sitting · one intake
- Battery scored blind to the self-rating; certification threshold held at the module default
- Fluency-performance association pre-registered; no items dropped post hoc

## Findings

Self-rated fluency and measured task score are effectively unrelated ( $r = 0.12$ ,  $n = 118$ ). The certification threshold admits high and low performers in near-equal proportion; what the self-assessment separates cleanly is module completion time, not task score.



Post-training self-rated fluency explains under 2% of the variance in measured task score ( $r = 0.12$ ,  $n = 118$ ); the fitted slope is indistinguishable from flat across the full confidence range.

## Implications

The self-assessment functions as a completion record wearing a capability label. Where an induction dashboard reports certified fluency, it is reporting who finished the module, not who can use the tools. The measure is stable, well-liked, and orthogonal to the thing it names.

Nothing in the instrument is broken: it records self-belief accurately and reliably. The gap is one of interpretation — a well-behaved measure of confidence has been read across the institution as a proxy for a competence it does not track.

Next: a task-anchored re-validation before the sign-off is trusted as a workforce-readiness indicator.

## Selected references

1. Baldwin, T. T., & Ford, J. K. (1988). Transfer of training: A review and directions for future research. *Personnel Psychology*, 41(1), 63–105. doi:10.1111/j.1744-6570.1988.tb00632.x
2. Saks, A. M., & Belcourt, M. (2006). An investigation of training activities and transfer of training in organizations. *Human Resource Management*, 45(4), 629–648. doi:10.1002/hrm.20135
3. Ng, W. (2012). Can we teach digital natives digital literacy? *Computers & Education*, 59(3), 1065–1078. doi:10.1016/j.compedu.2012.04.016
4. van Laar, E., van Deursen, A. J. A. M., van Dijk, J. A. G. M., & de Haan, J. (2017). The relation between 21st-century skills and digital skills: A systematic literature review. *Computers in Human Behavior*, 72, 577–588. doi:10.1016/j.chb.2017.03.010
5. Jin, Y., Martinez-Maldonado, R., Gašević, D., & Yan, L. (2024). GLAT: The Generative AI Literacy Assessment Test. arXiv:2411.00283.

De-identified task logs only; self-ratings and scores linked by a one-time code discarded at analysis. Battery and rubric pre-registered; the certification threshold was not adjusted for this study.

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